THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

Department of Software Engineering | Muzaffarabad

ACADEMIC LAB REPORT

LAB 02

Software Construction & Development

UML Activity & State Machine Diagrams

SEMESTER	**Fall 2026**
COURSE CODE	**SE-3102**
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REGISTRATION NO.	2024-SE-11

ASSIGNMENT SCOPE

Q1: Foodpanda activity diagram with swimlanes | Q2: Toaster state diagram

Q3: Elevator / Lift state machine diagram

QUESTION 1 | ACTIVITY DIAGRAM

Foodpanda Online Food Ordering - Swimlanes

Theory: A UML activity diagram shows the flow of work through actions and decisions. **Swimlanes** assign each action to the responsible participant: Customer, Foodpanda System, Payment Gateway, Restaurant, or Delivery Rider. Diamonds denote decisions/merges; the filled circle and bullseye indicate start and finish.

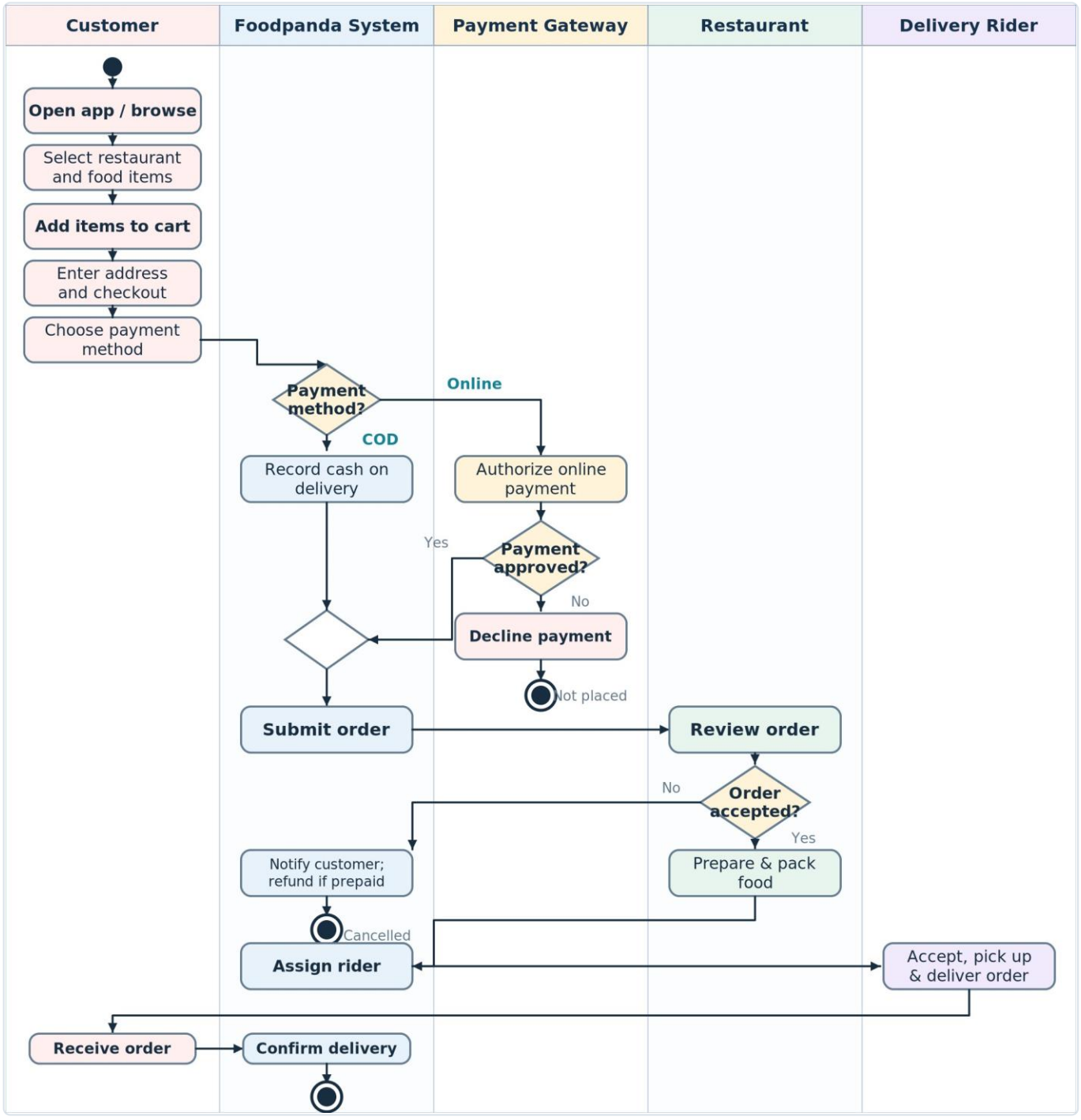


FIGURE 1 Foodpanda food-order activity diagram with five swimlanes.

Workflow: Browse and checkout → choose COD or online payment → submit order → restaurant acceptance → preparation and rider delivery → delivery confirmation. Payment failure or restaurant rejection ends the order.

QUESTION 2 | STATE DIAGRAM

Toaster - States, Activities, and Events

Theory: A UML state machine shows the states an object occupies and the events that change its state. The labels **entry /**, **do /**, and **exit /** describe what happens when a state starts, while it remains active, and when it ends.

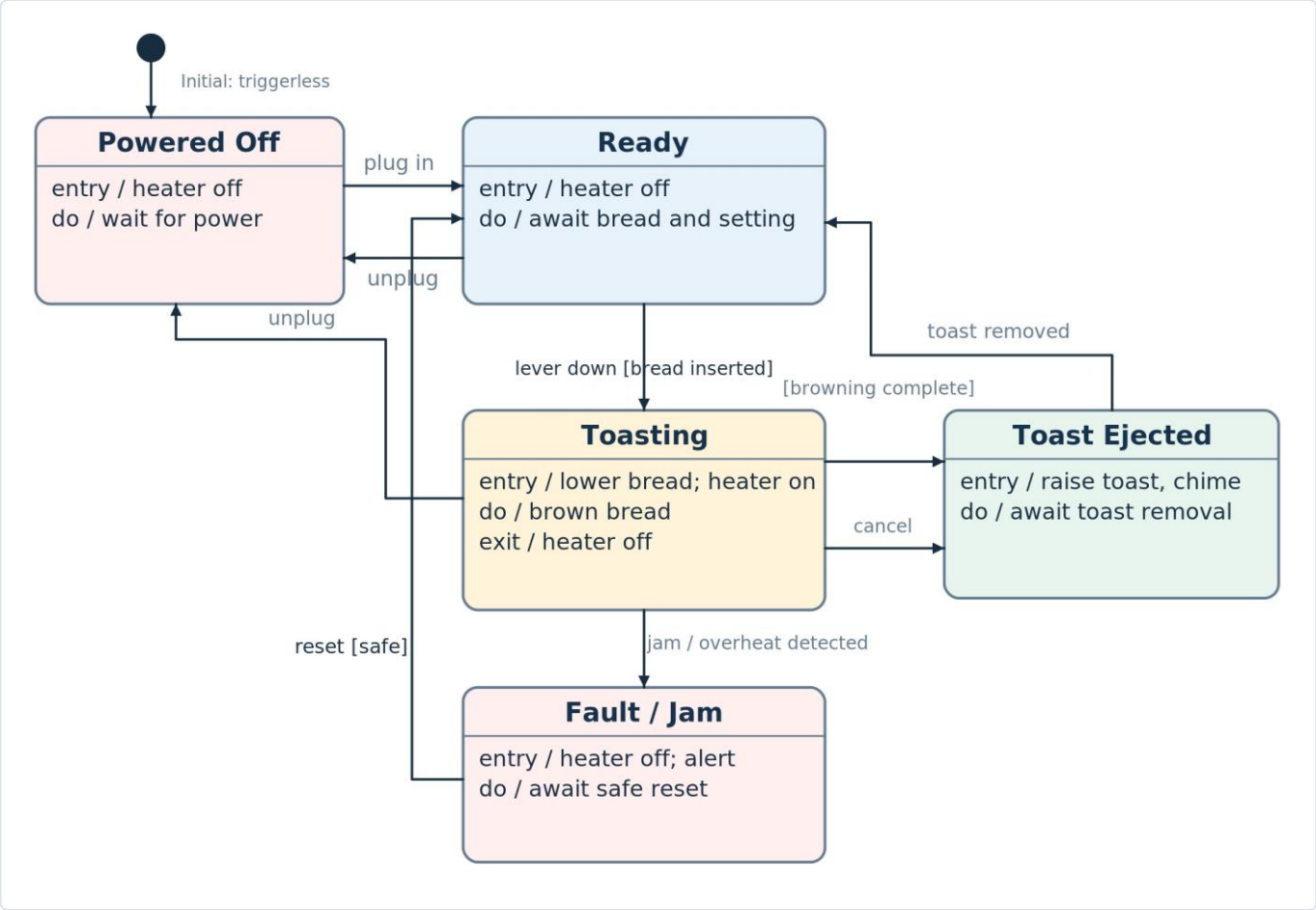


FIGURE 2 Toaster state machine with internal activities and guarded transitions.

Activities in each state

Powered Off: heater disabled. **Ready:** await bread and setting. **Toasting:** energize heater, brown bread, then switch heater off. **Toast Ejected:** pop up toast and wait for removal. **Fault/Jam:** turn heater off and await a safe reset.

Triggered vs. triggerless transitions

Triggered: plug in, unplug, lever down, cancel pressed, toast removed, jam/overheat detected, and reset.
Triggerless: the initial transition to Powered Off and the completion transition **[browning complete]** after the Toasting activity finishes.

QUESTION 3 | STATE MACHINE

Elevator / Lift - State Machine Diagram

Theory: The elevator moves only when a requested destination exists and its doors are fully closed. Its state machine represents direction of movement, door operations, user requests, obstruction handling, and the emergency-stop procedure.

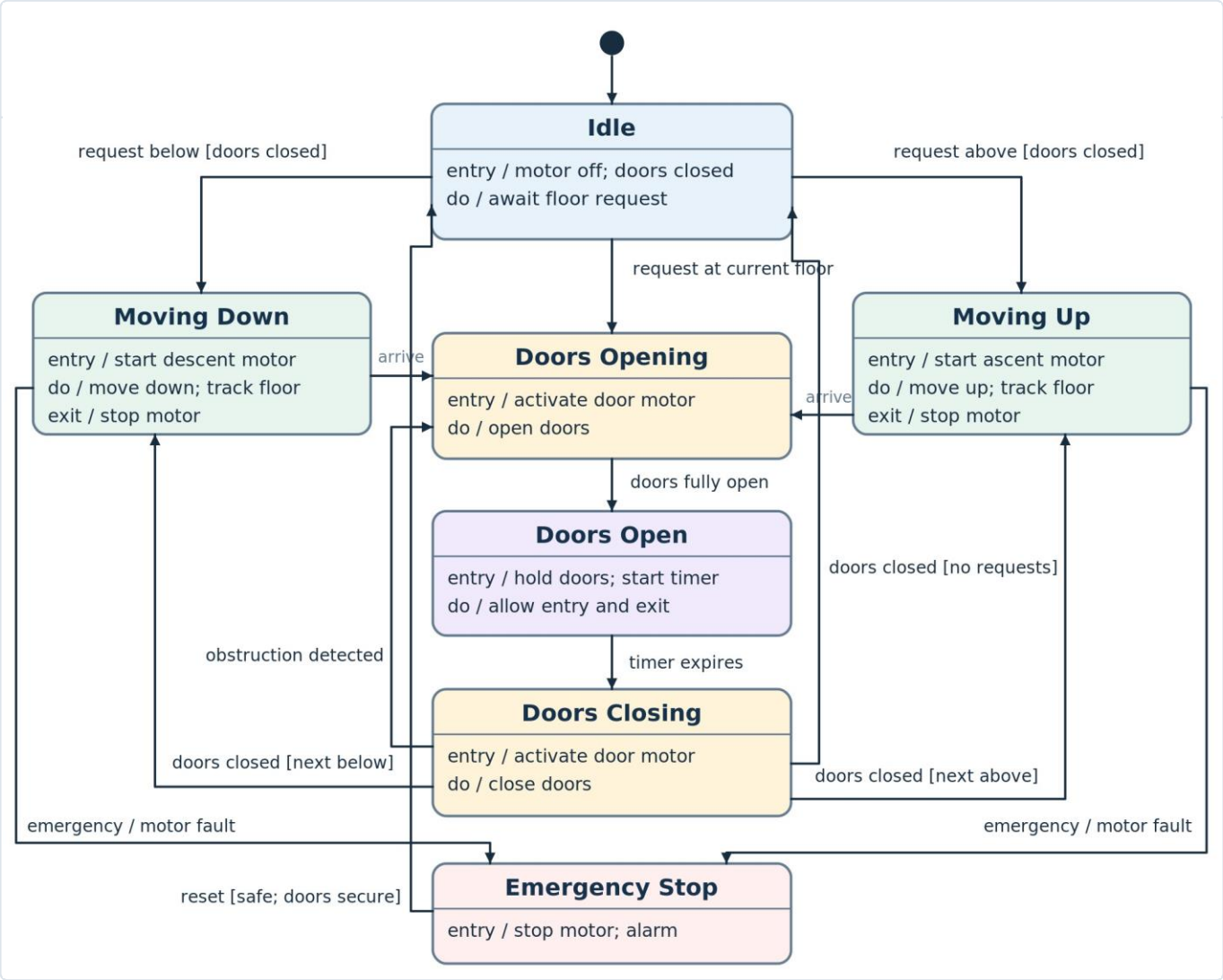


FIGURE 3 Elevator states, transitions, door interlocks, and emergency handling.

Interpretation

From **Idle**, a floor request starts **Moving Up**, **Moving Down**, or **Doors Opening**. Arrival opens the doors; after a timer expires, they close. An obstruction reopens them. Only after closing can the car move to a new destination or return to Idle. A motor fault/emergency stops the car until a safe reset.

Conclusion

Lab 2 demonstrates how UML behavioral diagrams clarify workflows and system lifecycles. The Foodpanda swimlane diagram assigns responsibilities across participants; the toaster and elevator models specify states, internal actions, events, guards, and safety-related transitions before implementation.